Poromechanical Modeling of Growth of Hydraulic Fracture Networks in Rocks with Pre-existing Weak Layers

Houlin Xu, Anh Nguyen, Yang Zhao, Zdeněk P. Bažant

SPONSOR: DOE, NSF

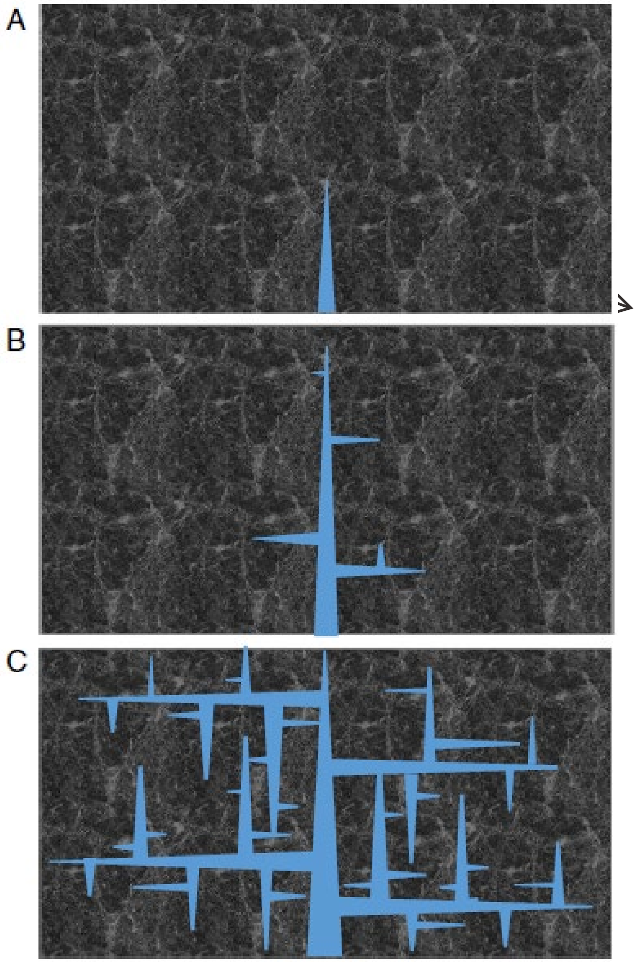
EMI 2025

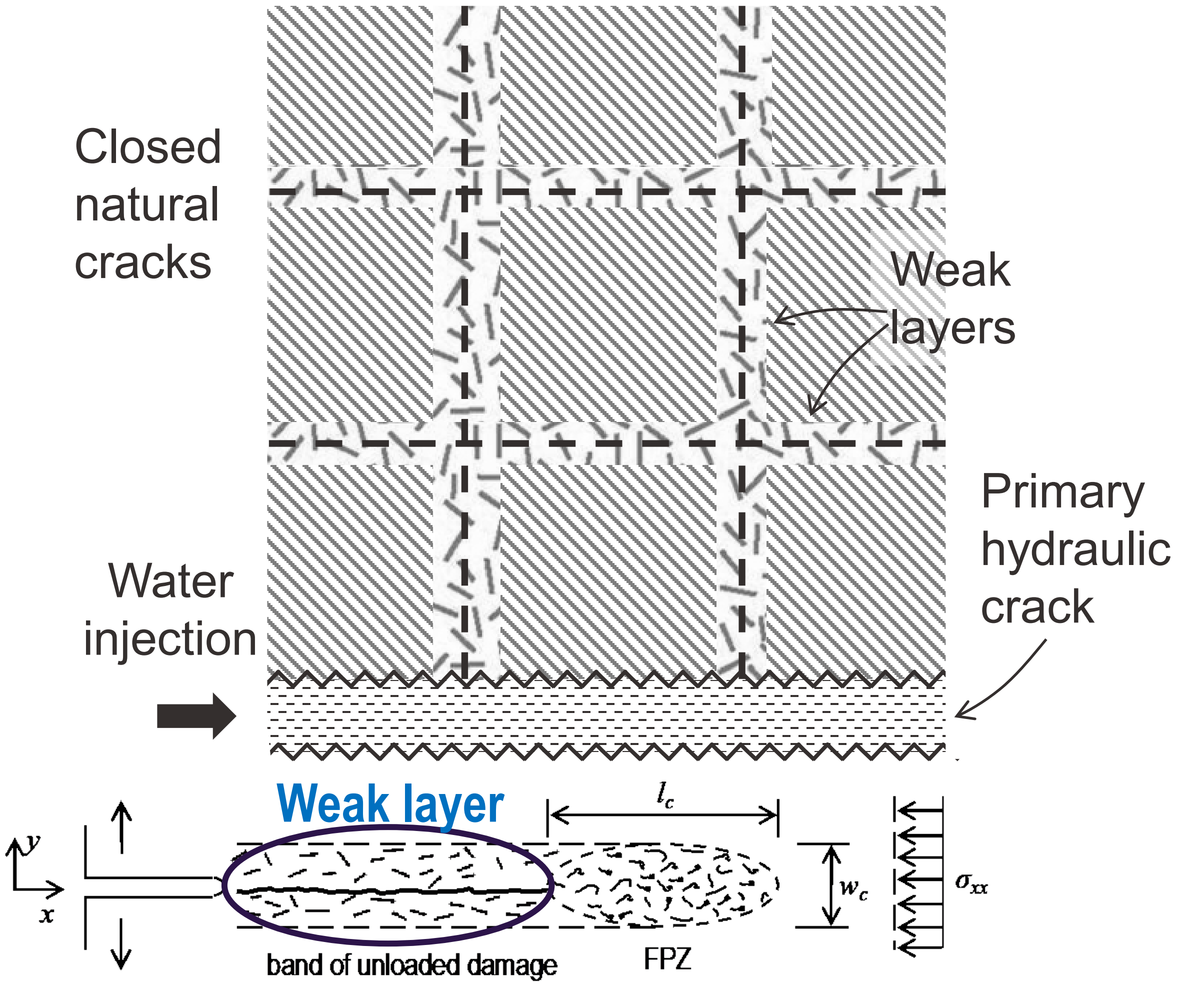
May 28, 2025

WEAK LAYERS that remain after creep closing of natural cracks formed in ~100 million years of geologic past. They are:

– **nano-cracked** (due to finite width of FPZ)

– **more permeable** than intact shale (even if cemented)





Solid-Fluid Stress Coupling: Biot's Poromechanics

Total stress for FEA:

$$\sigma = \sigma_s - b \cdot p$$

Solid stress via constitutive law

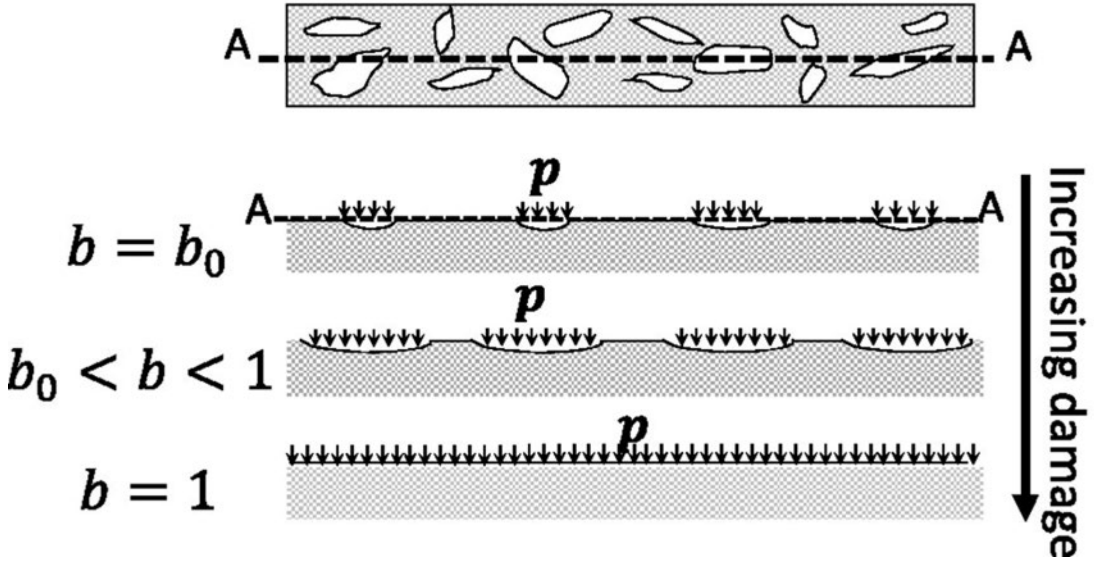
Stress from water pressure in pores and cracks

Due to oriented microcracking, **Biot coefficient** becomes a **tensor** (anisotropy)

$$b_{ij} = b_0\delta_{ij} + \beta\epsilon''_{ij}(\epsilon''_{kk}/3)^{-2/3}$$

$$\text{if } b_{kk} > 1 \text{ reset } \quad b_{ij} = \delta_{ij} \quad (\phi_0 \le b_0 \le 1)$$

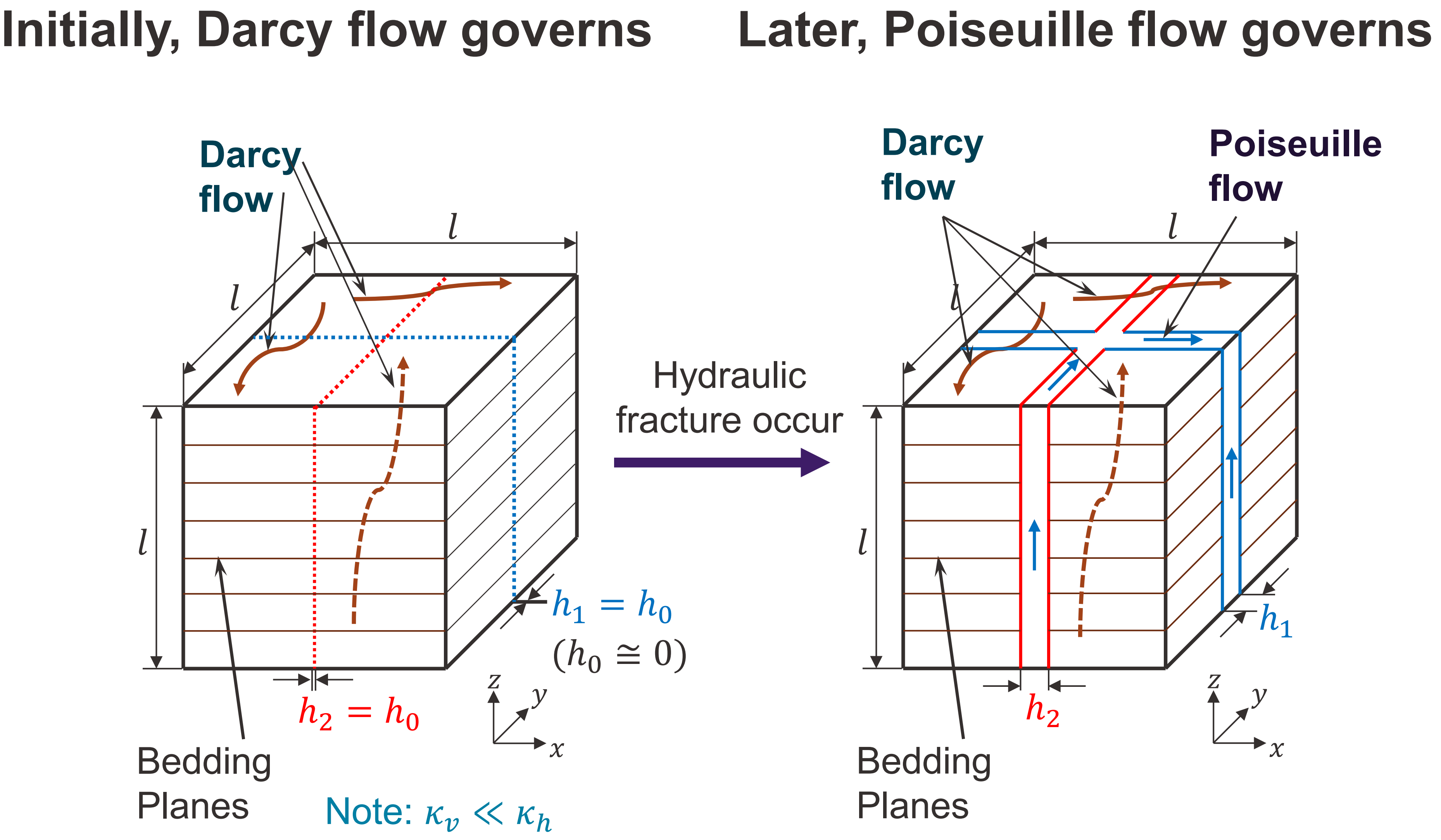
Stress and cracking damage in solid is obtained from the



Increase of Biot coefficient in transverse direction

Rahimi-Aghdam, S. *et al.* (2019). *Proceedings of the National Academy of Sciences*, *116*(5), 1532-1537.

Explanation of Lateral Crack Branching



where κ_v, κ_h = permeability of shale in vertical and horizontal direction

Governing equations

Mass conservation of the solid-fluid mixture:

$$\frac{1}{M}\dot{p} + b : \dot{\epsilon}^e + \mathbf{1} : \dot{\epsilon}^p + \nabla \cdot q = 0$$

Biot modulus

Darcy flux

Hydromechanical coupling parameters can be expressed as

$$\boldsymbol{b} = \mathbf{1} - \frac{\mathbf{1} : C}{3K_s}$$

$$\frac{1}{M} = \frac{\mathbf{1} : \boldsymbol{b} - 3\phi^f}{3K_s} + \frac{\phi^f}{K_f}$$

Balance of linear momentum

$$\nabla \cdot \sigma = \nabla \cdot (\sigma' - b\, p) = 0$$

Mathematical Analogy between Hydraulic Fracturing and Heat Conduction

Hydraulic fracturing

The mass flow rate

$$M_i = -k_{ij}\frac{\partial P}{\partial x_j} \qquad k_{ij} = \rho_0\left(1+\frac{P}{K}\right)B_{ij}$$

Mass conservation

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial M_i}{\partial x_i} = 0$$

$$\rho = \rho_0(1+P/K)$$

Governing equation

$$\frac{\rho_0 h}{K}\frac{\partial P}{\partial t} - \frac{\partial}{\partial x_i}\left[k_{ij}\frac{\partial P}{\partial x_j}\right] + \rho_0\left(1+\frac{P}{K}\right)\frac{\partial h}{\partial t} = 0$$

Boundary conditions

$$\begin{cases} P = \bar{P} & \text{at } \Gamma_1 \\ M_i n_i = -\bar{M} & \text{at } \Gamma_2 \\ M_i n_i = -\alpha(\bar{P}-P) & \text{at } \Gamma_3 \end{cases}$$

Initial condition

$$P = \bar{P}_0$$

Heat conduction

Fourier law

$$q_i = k_{ij}\frac{\partial T}{\partial x_j}$$

Thermal energy conservation

$$\rho c\frac{\partial T}{\partial t} + \frac{\partial q_i}{\partial x_i} - \rho Q = 0$$

Governing equation

$$\rho c\frac{\partial T}{\partial t} - \frac{\partial}{\partial x_i}\left[k_{ij}\frac{\partial T}{\partial x_j}\right] - \rho Q = 0$$

Boundary conditions

$$\begin{cases} T = \bar{T} & \text{at } \Gamma_1 \\ q_i n_i = -\bar{q} & \text{at } \Gamma_2 \\ q_i n_i = -\alpha(\bar{T}-T) & \text{at } \Gamma_3 \end{cases}$$

Initial condition

$$T = \bar{T}_0$$

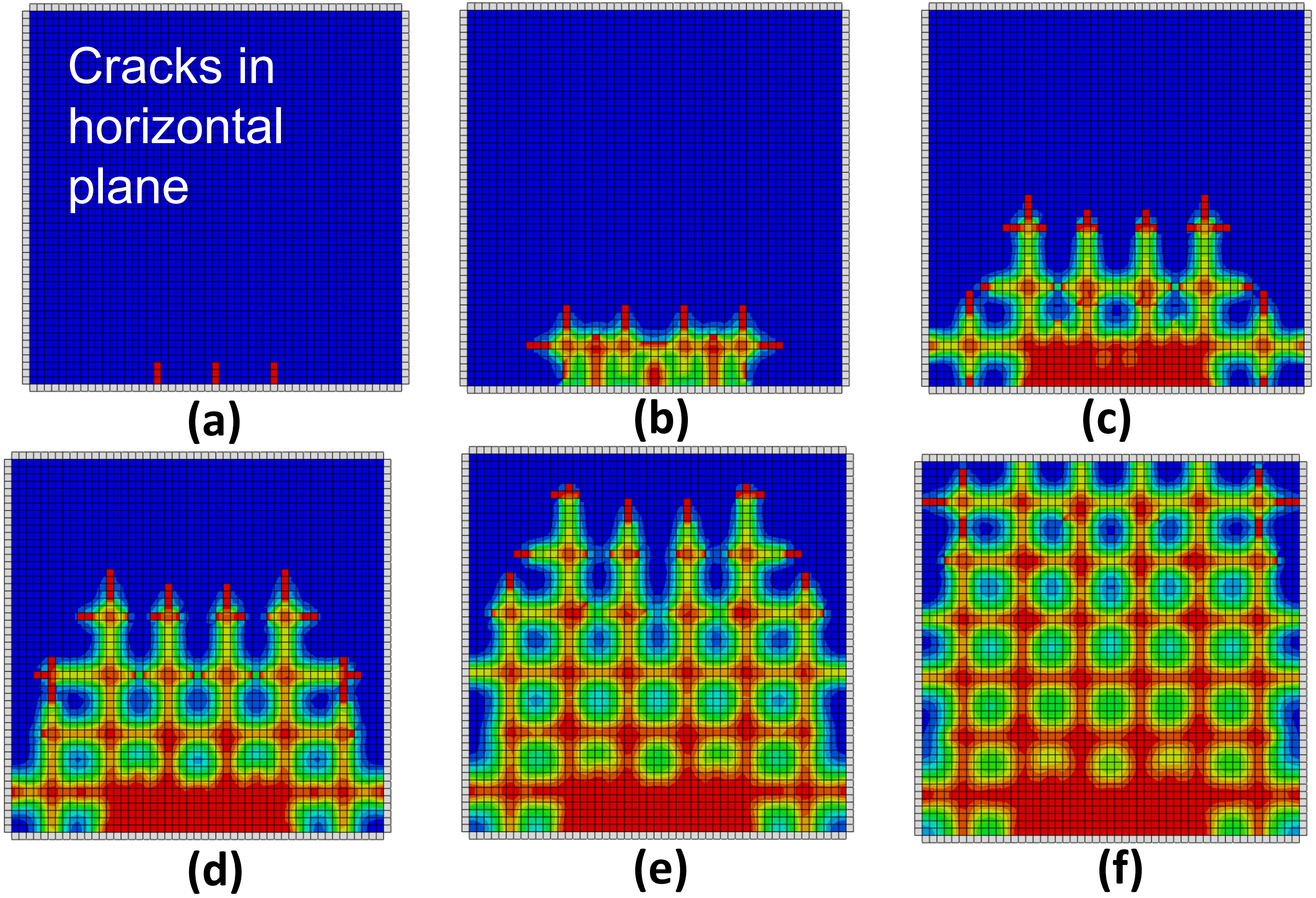
$$T \leftrightarrow P$$

$$q_i \leftrightarrow M_i$$

$$c = \frac{\rho_0 h}{\rho k}$$

$$Q = -\frac{\rho_0}{\rho}\left(1+\frac{P}{K}\right)\frac{\partial h}{\partial t}$$

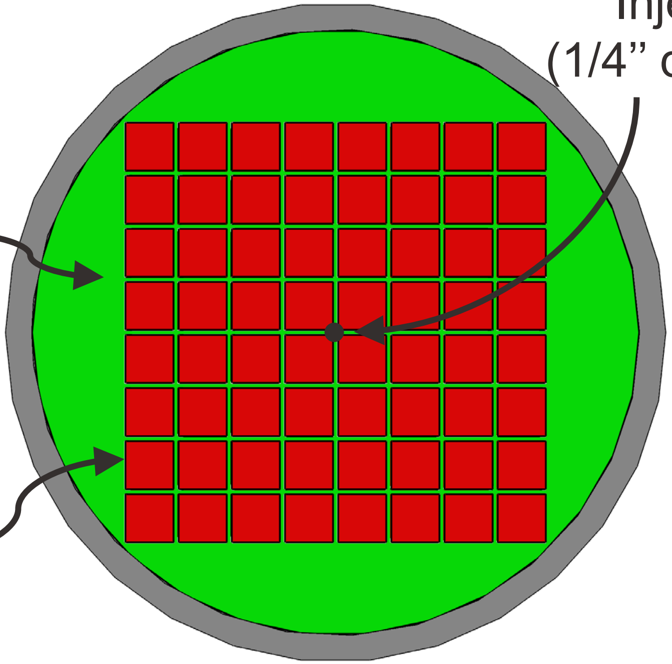
Previous simulation of theoretically feasible crack system evolution in fraccing of shale.

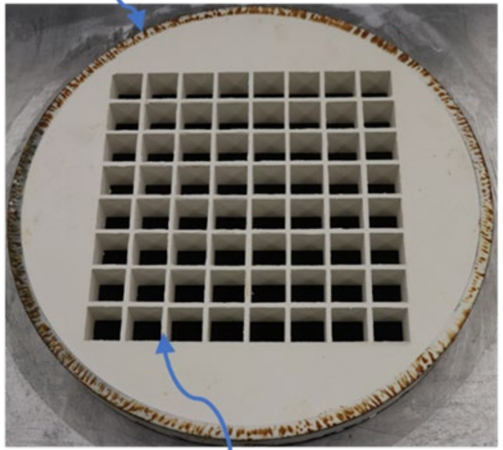


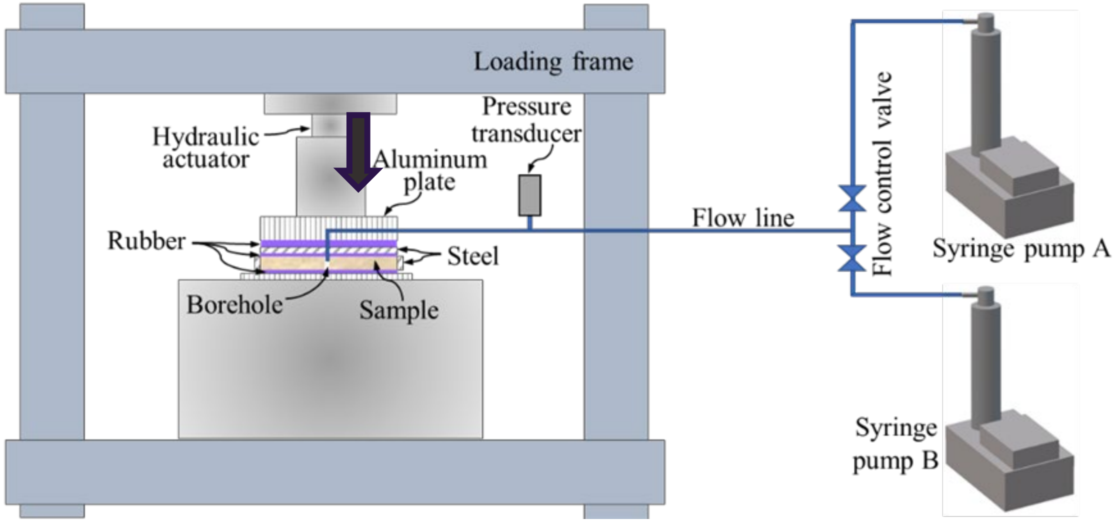
- 3 km deep, vertical crack faces are under high tectonic and overburden crack-parallel stresses σ_{xx} , σ_{zz} , combined by body forces applied by diffusion pressure gradients.
- Effects of σ_{xx}, σ_{zz} in fracking of shale are significant
- Biot coefficient anisotropy was found to be essential

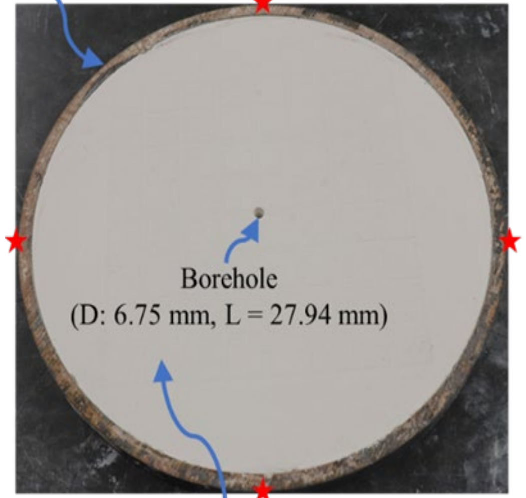
S. Rahimi-Aghdam - ZP Bazant (2019). PNAS 116 (5), 1532—1537.

Experimental results on a plaster prototype



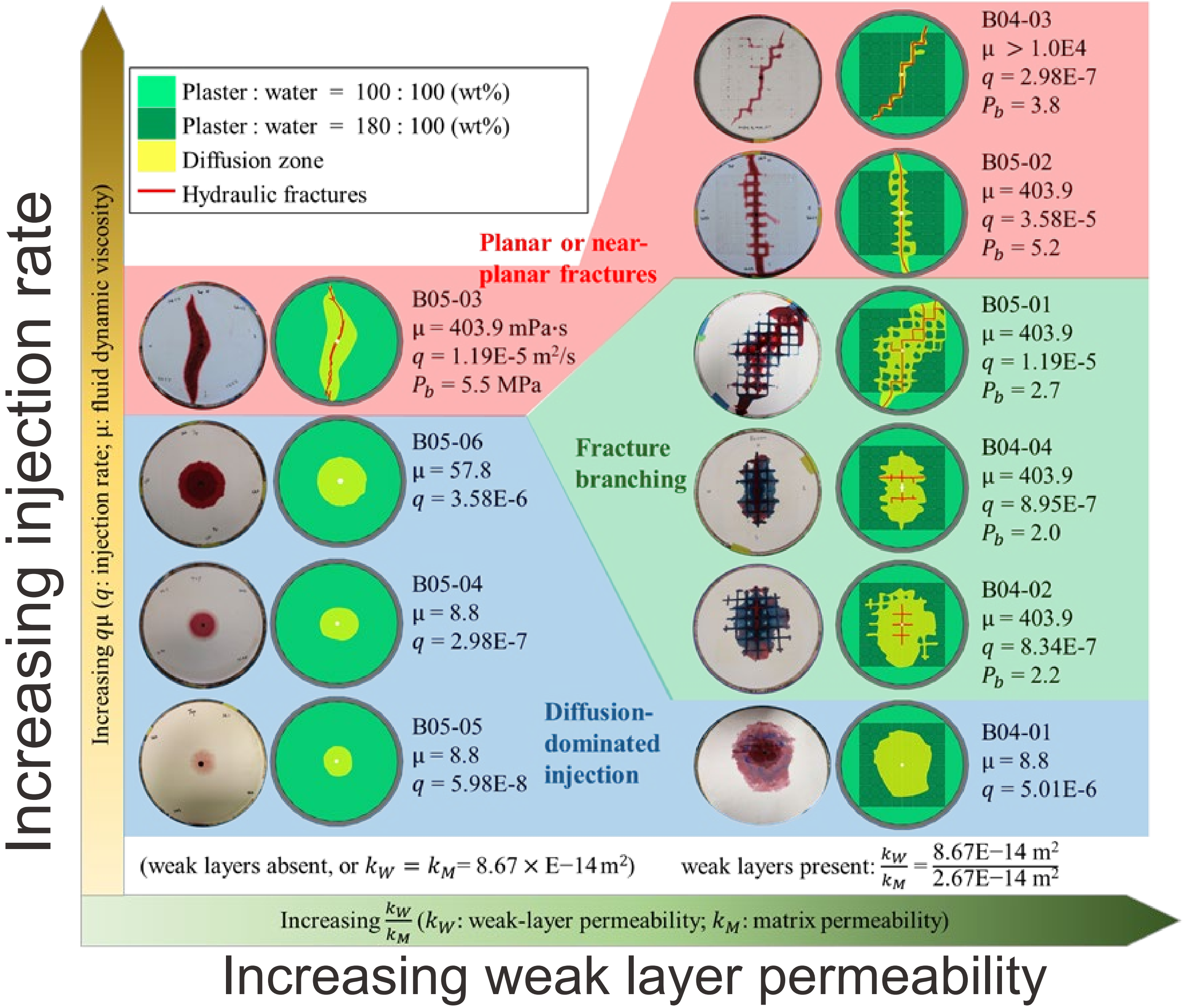






Wenfeng Li et al. (2021), Geophysical Research Letters 48 (12), e2021GL093321, pp. 1—12 (LA-UR-21-21928)

Tests to demonstrate crack branching in plaster



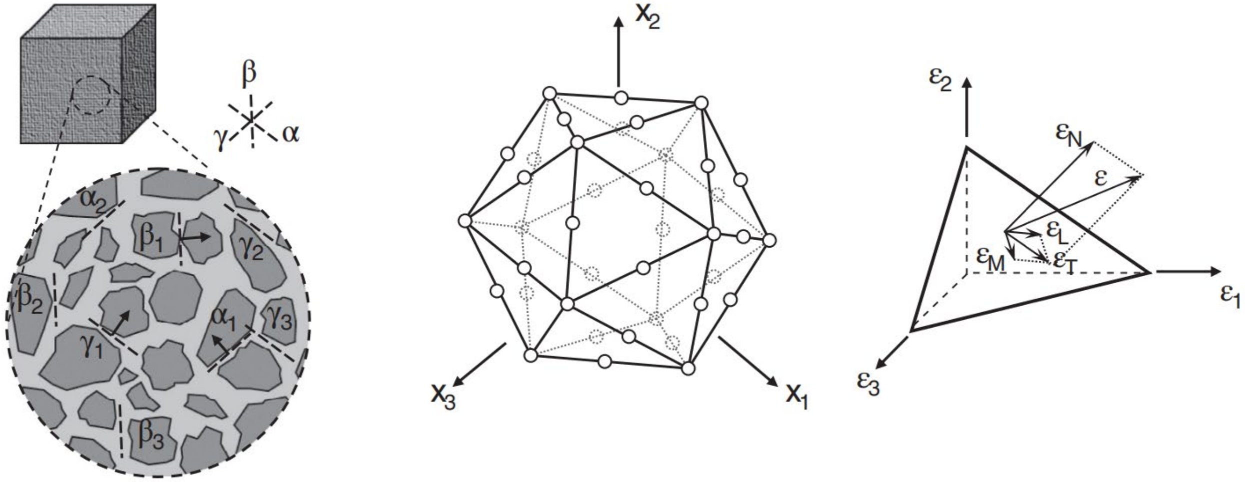
Wenfeng Li et al. (2021), Geophysical Research Letters 48 (12), e2021GL093321, pp. 1—12 (LA-UR-21-21928)

Constitutive Law and Evolution of Biot Tensor

Biot effective stress

$$\sigma' = \mathbb{C}^{\mathrm{e}} : \epsilon^{e}$$

$$b_{ij} = \delta_{ij} - \frac{\delta_{kl} C^{e}_{ijkl}}{3K_s} = \delta_{ij} - \frac{C^{e}_{ijkk}}{3K_s}$$



Principle of Virtual Work

$$\delta\sigma'_{ij} = \mathbb{C}^{e}_{ijkl} : \epsilon^{e}_{kl} = \frac{3}{2\pi}\int_{\Omega}\left(N_{ij}\delta\sigma^{e}_{N} + M_{ij}\delta\sigma^{e}_{M} + L_{ij}\delta\sigma^{e}_{L}\right)\mathrm{d}\Omega$$

where

$$\delta\sigma^{e}_{N} = E_N \delta\epsilon^{e}_{N} = E_N N_{kl}\delta\epsilon^{e}_{kl}$$
$$\delta\sigma^{e}_{L} = E_T \delta\epsilon^{e}_{L} = E_T M_{kl}\delta\epsilon^{e}_{kl}$$
$$\delta\sigma^{e}_{M} = E_T \delta\epsilon^{e}_{M} = E_T L_{kl}\delta\epsilon^{e}_{kl}$$

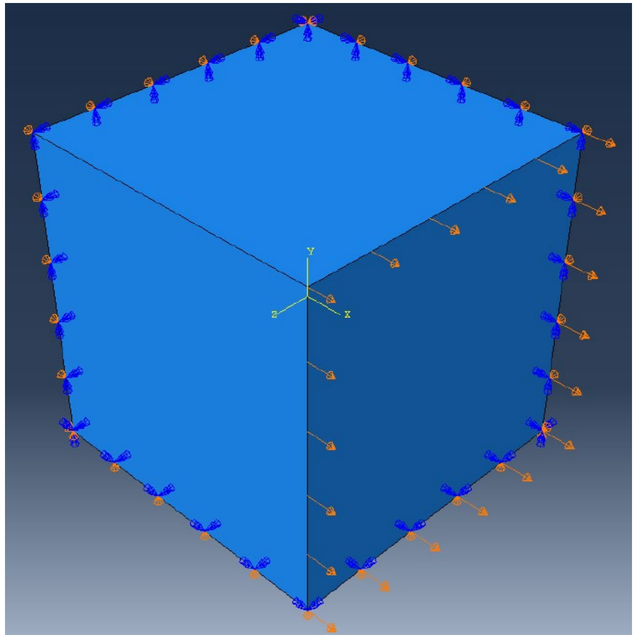
$$\Downarrow$$

$$\delta\sigma'_{ij} = \frac{3}{2\pi}\int_{\Omega}\left(E_N N_{ij} N_{kl}\delta\epsilon^{e}_{kl} + E_T M_{ij} M_{kl}\delta\epsilon^{e}_{kl} + E_T L_{ij} L_{kl}\delta\epsilon^{e}_{kl}\right)\mathrm{d}\Omega$$

Constitutive Law and Evolution of Biot Tensor

$$\mathbb{C}^e = \frac{3}{2\pi}\int_\Omega (E_N N \otimes N + E_T M \otimes M + E_T L \otimes L) d\Omega$$
$$\approx 6 \sum_{i=1}^{N_m} w_i \left(E_N^i N^i \otimes N^i + E_T^i M^i \otimes M^i + E_T^i L^i \otimes L^i\right)$$

$N_m = 37,\ \Sigma_\mu w_\mu = \frac{1}{2}$

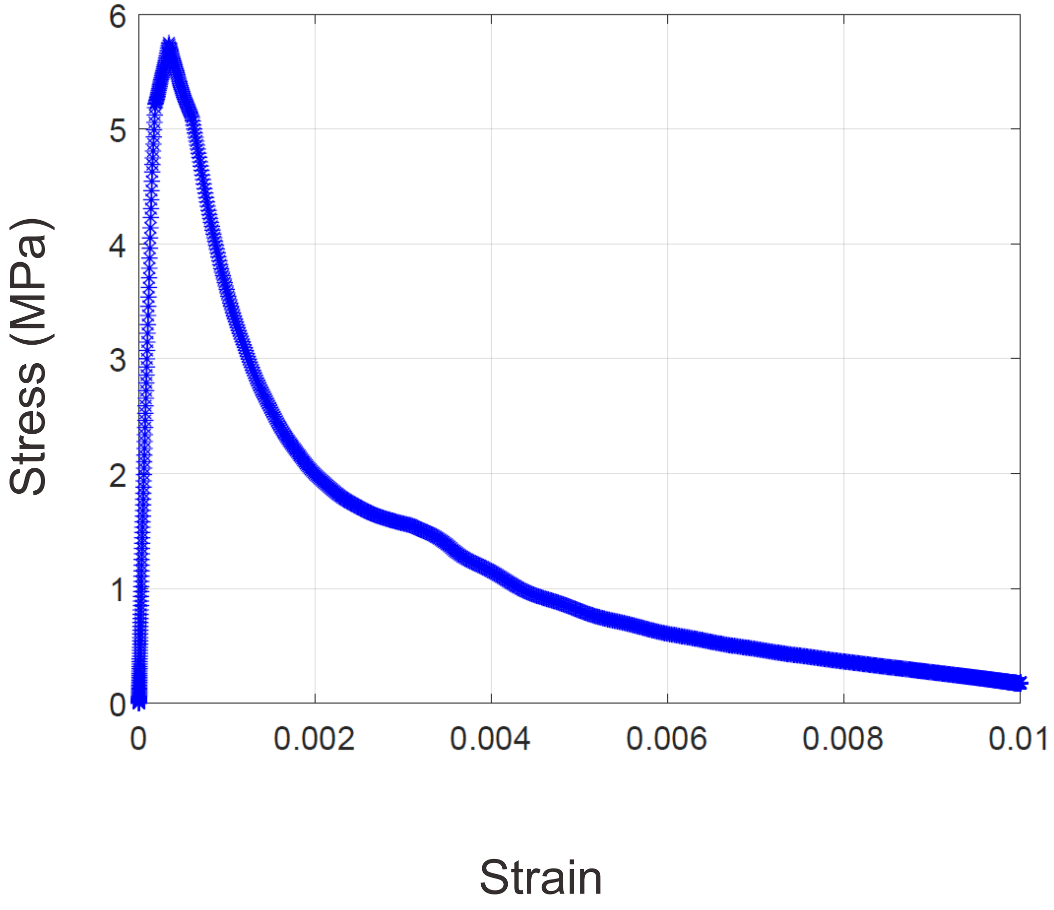


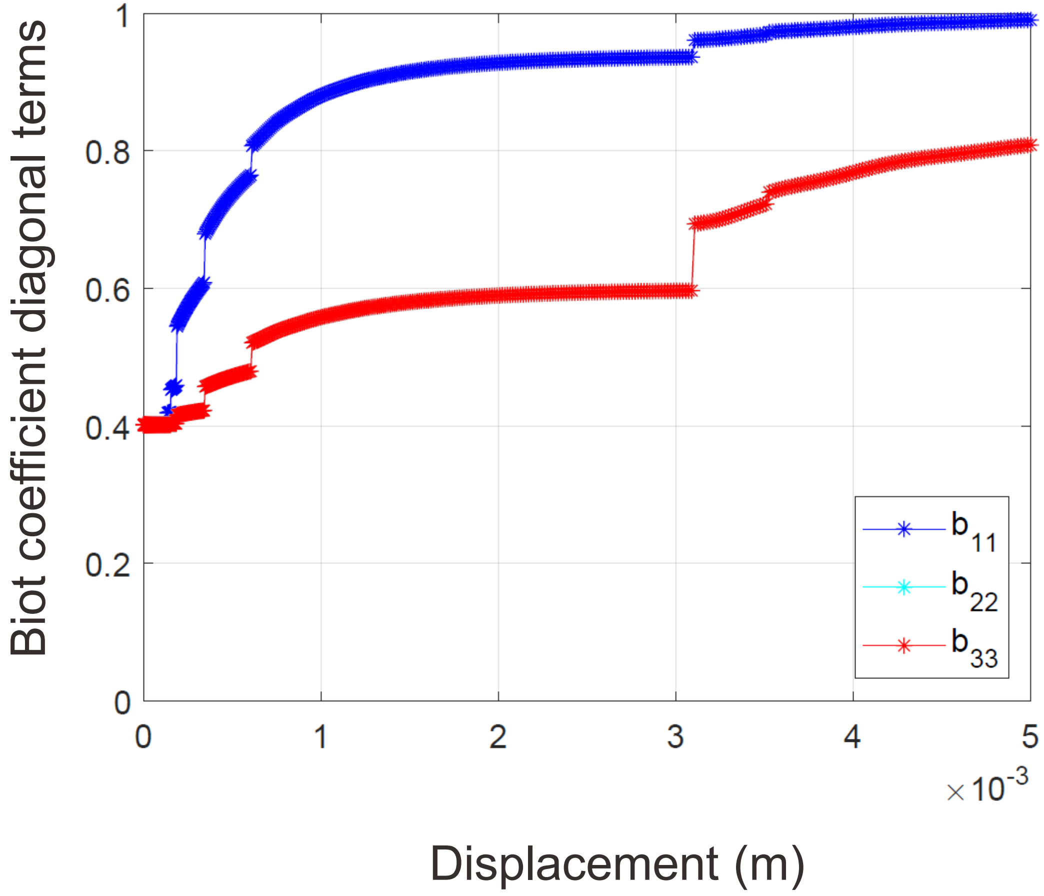
Empirical $b_{ij} = b_0 \delta_{ij} + \beta \epsilon''_{ij} \left(\frac{\epsilon''_{kk}}{3}\right)^{-2/3}$

$$\Downarrow$$

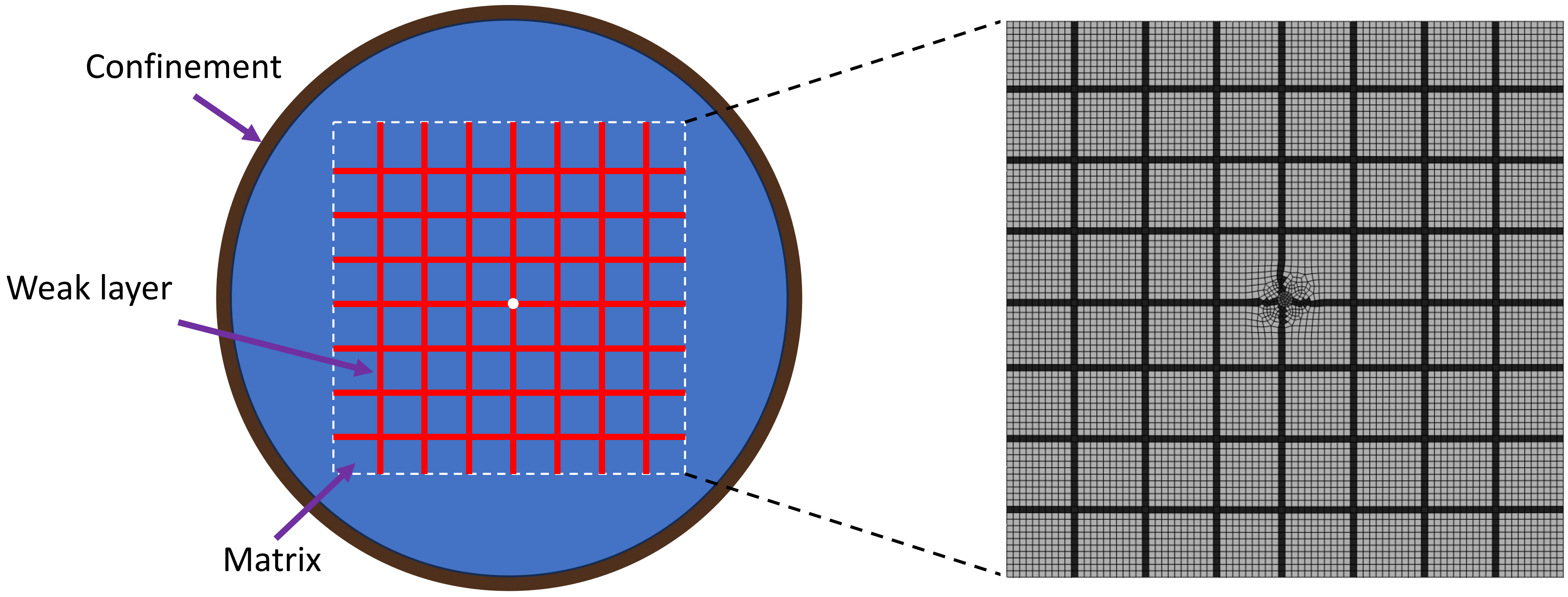
Analytical $\boldsymbol{b} = 1 - \frac{2 \sum_{i=1}^{N_m} w_i \left(E_N^i \boldsymbol{N}^i + E_T^i \boldsymbol{M}^i + E_T^i \boldsymbol{L}^i\right)}{K_s}$

Evolution of Biot Tensor and Verification





FEM Simulation



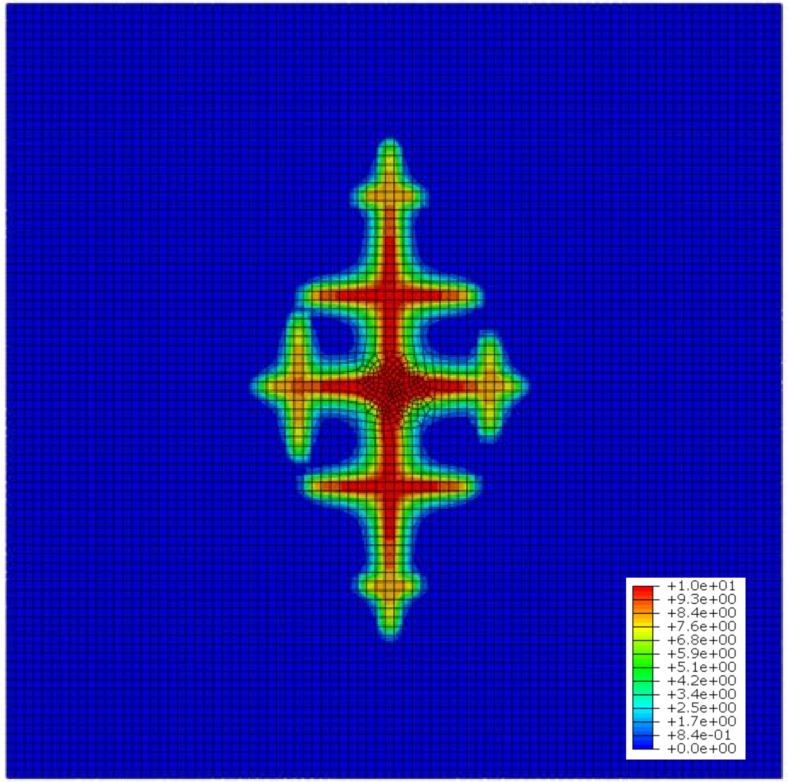
Pore pressure distributions with different flow rate

High flow rate

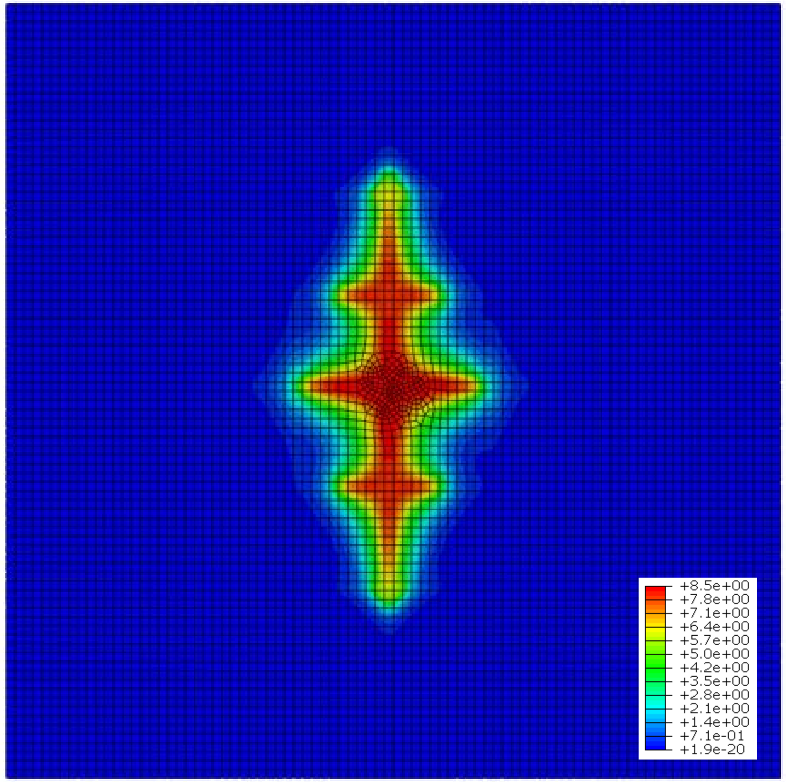
Low flow rate

Fracture Propagates

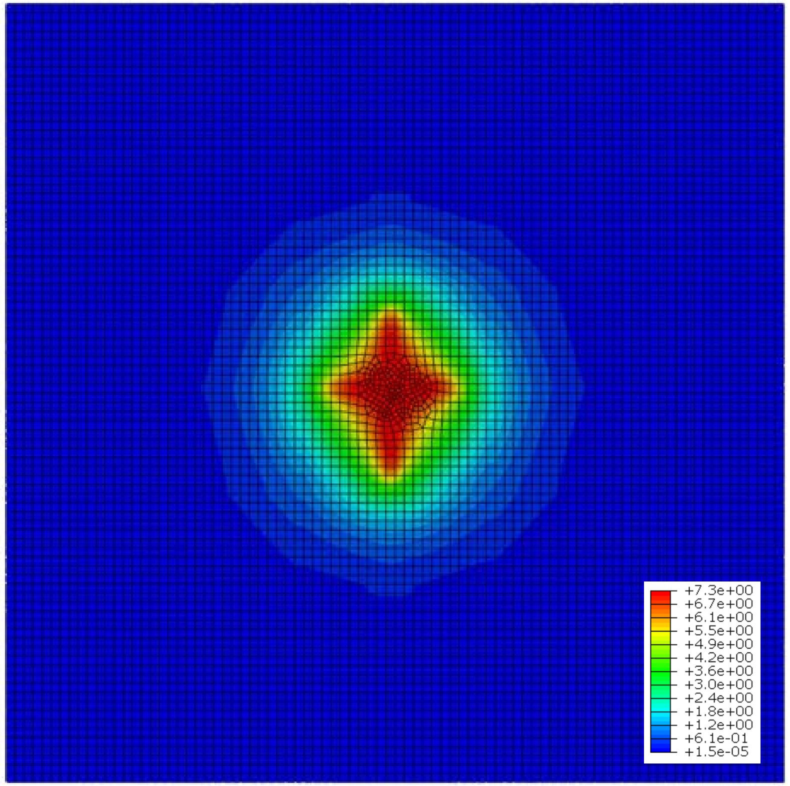
Diffusion Dominant



$q_{in} = 1 \times 10^{-6}\ \mathrm{m}^3/s$

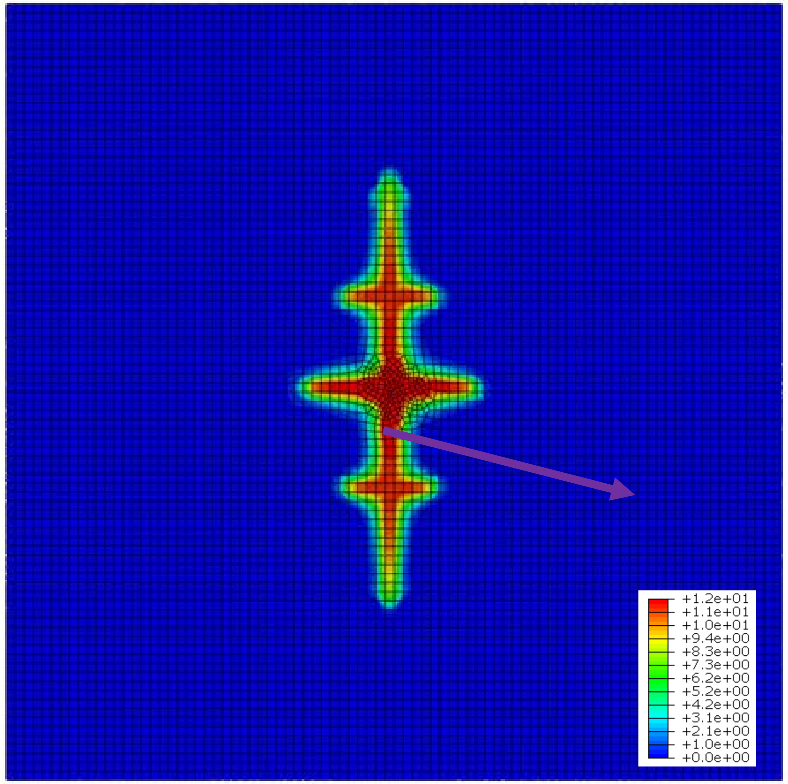


$q_{in} = 1.7 \times 10^{-7}\ \mathrm{m}^3/s$

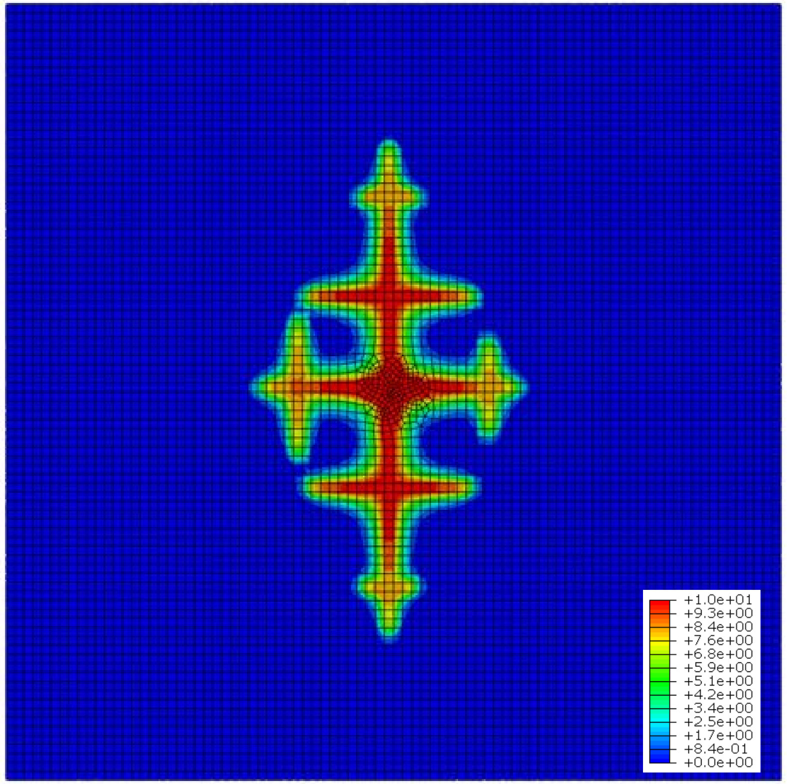


$q_{in} = 2.3 \times 10^{-8}\ \mathrm{m}^3/s$

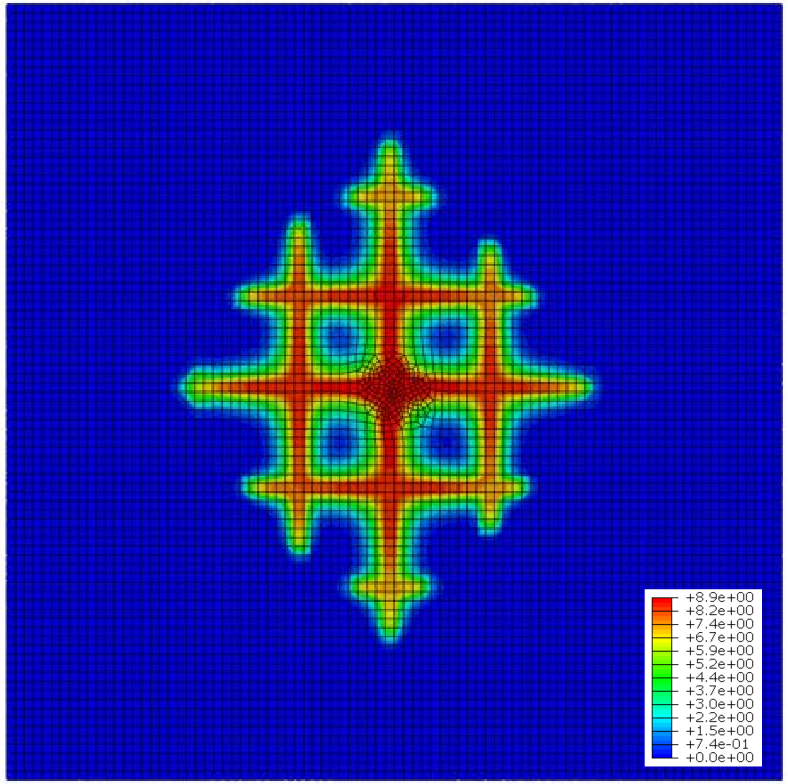
Pore pressure distributions with different initial biot coefficient



$\bar{b} = 0.6$



$\bar{b} = 0.8$



$\bar{b} = 1.0$

Conclusions

- **Biot Tensor Evolution**: The evolution of the Biot tensor calculated using microplane model degradation successfully captures the anisotropic and progressive failure processes.
- **Model Validation**: The proposed poromechanical model successfully reproduces the LANL laboratory-scale hydraulic fracturing experiments on synthesized rock specimens.
- **Fracture Pattern Classification**: Two distinct hydraulic fracturing patterns were identified: formation of one dominant planar crack versus development of complex fracture networks.
- **Injection Rate Control**: Higher fluid injection rates create greater pore pressure contrasts between rock matrix and weak layers, facilitating fracture propagation and branching.

Thank you!

Northwestern | McCORMICK SCHOOL OF ENGINEERING